MIL-I-23053/1B
01 May 1986
SUPERSEDING
MIL-I-23053/1A
20 May 1976

MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYCHLOROPRENE, FLEXIBLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range:

Class 1: -55°C (-67°F) to +90°C (+194°F)
Class 2: -70°C (-94°F) to +121°C (+250°F)

Color: The sleeving shall be furnished in a black color that conforms to Class II of MIL-STD-104 (see 1.2.2 and 3.4.1.5).

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Normal operating temperature
Class 2 - Extended operating temperature

NOTICE:

Class 1 sleeving because of its temperature limitations should not be considered for new design applications. Class 2 sleeving can be substituted for Class 1 sleeving.

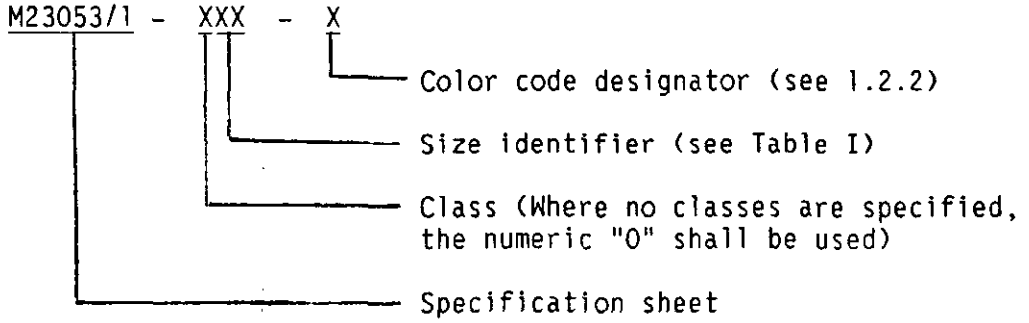
Longitudinal change: +1, -10 percent

NOTE: Entire specification sheet revised.

AMSC N/A FSC 5970
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Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers shown as follows:



Example: Class 2, black, 1.000 inch (25.4 mm) as supplied ID sleeving shall be identified as M23053/1-207-0.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/1-101-0	.250 (6.4)	.143 (3.6)	.035 ± .010 (.89 ± .25)
M23053/1-102-0	.375 (9.5)	.211 (5.4)	.040 ± .010 (1.01 ± .25)
M23053/1-103-0	.500 (12.7)	.286 (7.3)	.048 ± .015 (1.21 ± .38)
M23053/1-104-0	.625 (15.9)	.357 (9.1)	.052 ± .015 (1.32 ± .38)
M23053/1-105-0	.750 (19.1)	.428 (10.9)	.057 ± .015 (1.44 ± .38)
M23053/1-106-0	.875 (22.2)	.500 (12.7)	.065 ± .020 (1.65 ± .51)
M23053/1-107-0	1.000 (25.4)	.570 (14.5)	.070 ± .020 (1.77 ± .51)
M23053/1-108-0	1.250 (31.8)	.714 (18.1)	.087 ± .020 (2.20 ± .51)
M23053/1-109-0	1.500 (38.1)	.857 (21.8)	.095 ± .020 (2.41 ± .51)
M23053/1-110-0	1.750 (44.5)	1.000 (25.4)	.107 ± .020 (2.71 ± .51)
M23053/1-111-0	2.000 (50.8)	1.140 (29.0)	.110 ± .020 (2.79 ± .51)
M23053/1-112-0	3.000 (76.2)	1.710 (43.4)	.125 ± .020 (3.17 ± .51)
M23053/1-113-0	4.000 (101.6)	2.280 (57.9)	.140 ± .020 (3.55 ± .51)
Class 2			
M23053/1-201-0	.250 (6.4)	.143 (3.6)	.035 ± .010 (.89 ± .25)
M23053/1-202-0	.375 (9.5)	.211 (5.4)	.040 ± .010 (1.01 ± .25)
M23053/1-203-0	.500 (12.7)	.286 (7.3)	.048 ± .015 (1.21 ± .38)
M23053/1-204-0	.625 (15.9)	.357 (9.1)	.052 ± .015 (1.32 ± .38)
M23053/1-205-0	.750 (19.1)	.428 (10.9)	.057 ± .015 (1.44 ± .38)
M23053/1-206-0	.875 (22.2)	.500 (12.7)	.065 ± .020 (1.65 ± .51)
M23053/1-207-0	1.000 (25.4)	.570 (14.5)	.070 ± .020 (1.77 ± .51)
M23053/1-208-0	1.250 (31.8)	.714 (18.1)	.087 ± .020 (2.20 ± .51)
M23053/1-209-0	1.500 (38.1)	.857 (21.8)	.095 ± .020 (2.41 ± .51)
M23053/1-210-0	1.750 (44.5)	1.000 (25.4)	.107 ± .020 (2.71 ± .51)
M23053/1-211-0	2.000 (50.8)	1.140 (29.0)	.110 ± .020 (2.79 ± .51)
M23053/1-212-0	3.000 (76.2)	1.710 (43.4)	.125 ± .020 (3.17 ± .51)
M23053/1-213-0	4.000 (101.6)	2.280 (57.9)	.140 ± .020 (3.55 ± .51)

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1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The color code identified is the standard acquisition color.

Unrestricted shrinkage procedures: Test method 4.5.5; 175° ± 2°C (347° ± 4°F) for 10 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and condition
As supplied:		
ID, minimum	Table I	4.6.3
Low temperature flexibility	No cracks	4.6.7.1 Class 1:-55° ± 2°C (-67° ± 4°F); Class 2: -70° ± 2°C (-94° ± 4°F)
Heat shock	No cracks, flowing or dripping	4.6.8 Class 1: 150° ± 2°C; (302° ± 4°F); Class 2: 200° ± 2°C (392° ± 4°F)
Restricted shrinkage	No cracks	4.6.6.1.1 135° ± 2°C (275 ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1,500 (10.3)	4.6.14 ASTM D 412
Ultimate elongation, percent, minimum	225	4.6.14 ASTM D 412

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and condition
Tensile stress, at 200 percent elongation psi (MPa), maximum	1,500 (10.3)	4.6.13.2 ASTM D 412
Dielectric strength, volts/mil (Kv/mm), minimum	300 (76.2)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-CM, minimum	1 x 10^{11}	4.6.2 ASTM D 876
Corrosion	No pitting or corrosion	4.6.10.1 and 4.6.10.2 at 150° ± 2°C (302° ± 4°F) for 16 hours.
Water absorption, percent, maximum	1.0	4.6.2 ASTM D 570, 24 hrs at 23°C
Heat resistance, properties after:		4.6.9 Class 1: 100° ± 2°C (212° ± 4°F), 96 hours; Class 2: 121° ± 2°C (250° ± 4°F), 168 hours
Tensile strength, psi (MPa), minimum	1,200	
Ultimate elongation, percent, minimum	Class 1 - 150 Class 2 - 175	
Dielectric strength, volt/mil (Kv/mm), minimum	300 (11.8)	
Fluid resistance, properties after:		4.6.11
Tensile strength, psi (MPa), minimum	Class 1 - 900 (6.2) Class 2 - 1,000 (6.9)	
Ultimate elongation, percent, minimum	Class 1 - 150 Class 2 - 175	

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and condition
Dielectric strength, volts/mil (Kv/mm), minimum	Class 1 - 200 (7.9) Class 2 - 250 (9.8)	
Flammability	Self extinguishing 15 seconds; 3 inches	4.6.15.2
Fungus resistance	No growth	4.6.2 ASTM G 21

1/ Unless otherwise specified, requirements, test procedures and conditions are for all classes.

Shelf life conditions: Supplier shall certify to storage at 18 to 35°C (65 to 95°F) for 1 year. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable flexible polychloroprene sleeving is intended for use on cables or harness systems in equipment that is subject to high levels of physical abuse.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - EA, MI
Navy - EC
Air Force - 99
DLA - GS

Preparing activity:
Navy - AS

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